DSM A2 Part 2 Final Report: MongoDB + Neo4j Querying

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Contents

1 MongoDB Querying

Workflow and Execution

The full MongoDB Part 2 pipeline is implemented in `queries/p2_mongo.py`. It writes complete aggregation pipelines, executes all required queries, saves tabular outputs, and generates visualizations.

Run sequence:

```
source .venv/bin/activate
pip install -r requirements.txt
python queries/p2_mongo.py
```

Generated artifacts:

- Pipelines: `queries/mongo_part2_queries.txt`
- Data outputs: `report/p2/data/q*.csv`, `report/p2/data/q*.json`
- Figures: `report/p2/figures/q*.png`

Q1: Cohort Analysis of User Reviewing Behaviour

Cohorts are defined by year of `users.yelping_since`. For each cohort, the query computes mean and standard deviation of stars, mean review character length, mean useful votes per review, and star-bucket proportions.

Key outputs from `q1_summary.json`:

- Highest mean star cohort: **2005** (**3.903**)
- Highest mean useful-votes cohort: **2004** (**2.276**)
- Total cohorts: **19**

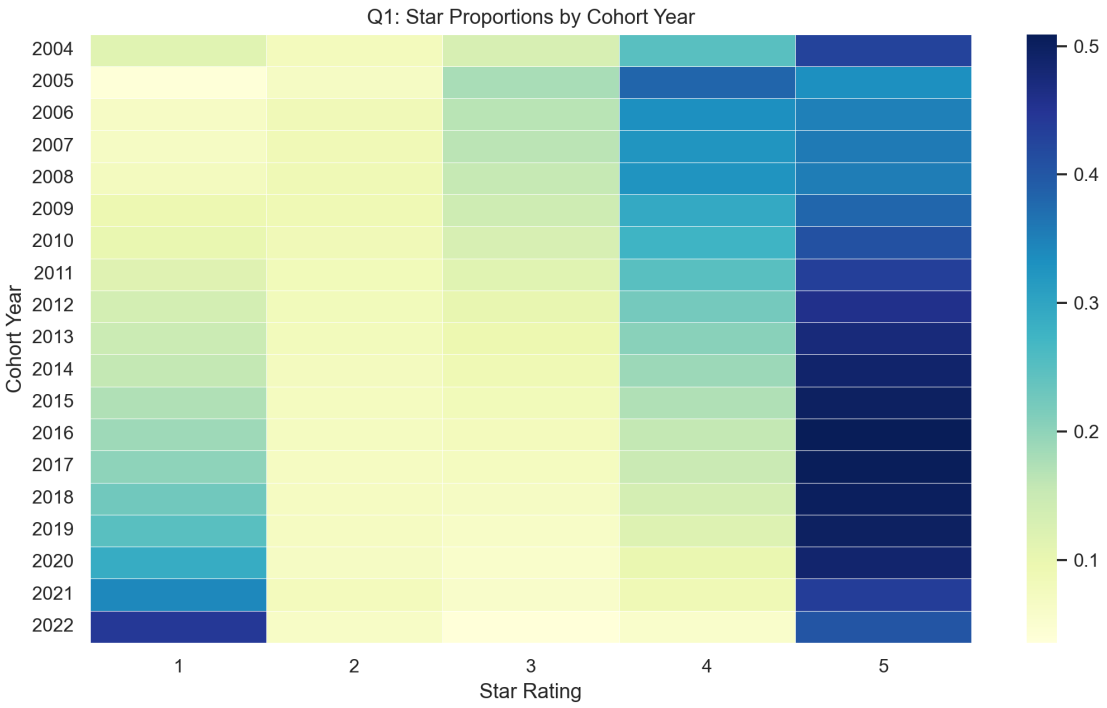

Figure 1: Q1 heatmap: star proportions by cohort year.

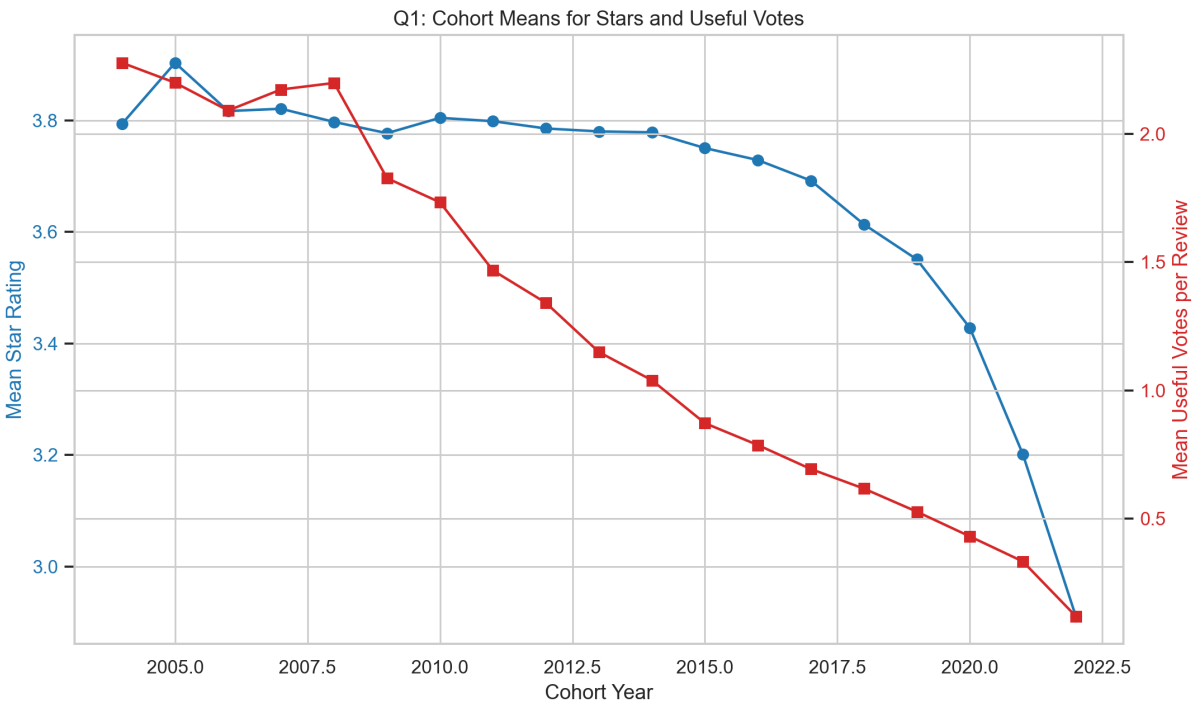

Figure 2: Q1 line plot: mean stars and mean useful votes by cohort year.

Interpretation:

- Older cohorts show stronger useful-vote and rating signals than recent cohorts.
- Later cohorts (notably post-2018) trend lower on both dimensions.

Q2: Month-over-Month Trend Consistency by Category

For each category with at least 500 reviews, monthly average stars are computed and consistency is measured as the proportion of consecutive month-pairs with increase/decrease.

Top 3 upward categories:

- Japanese Curry (0.544)
- Home Cleaning (0.543)
- Beer Gardens (0.541)

Top 3 downward categories:

- Medical Centers (0.566)
- Walk-in Clinics (0.554)
- Eyelash Service (0.550)

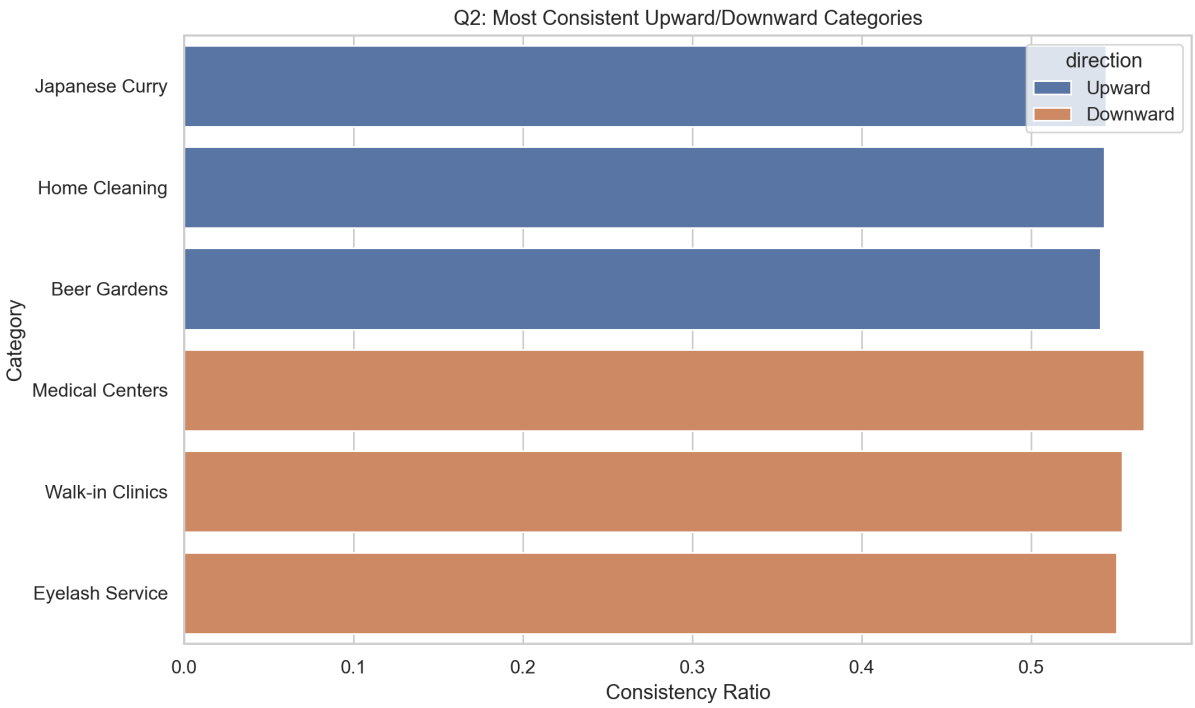

Figure 3: Q2 bar plot: consistency ratios for most upward/downward categories.

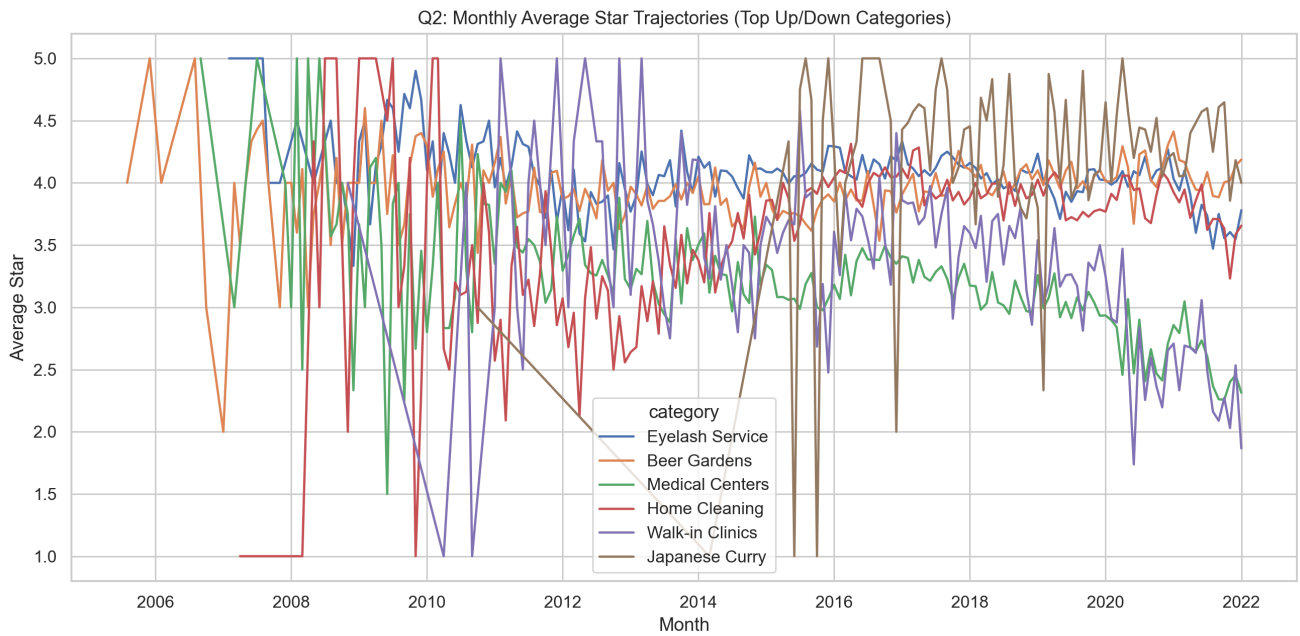

Figure 4: Q2 line plot: monthly trajectories for selected categories.

Interpretation:

- Directional consistency exists, but no category is fully monotonic.
- Both upward and downward groups show meaningful month-to-month volatility.

Q3: Check-in Class Cross-tabulation

Businesses are classified into low/medium/high check-in classes using quartiles from check-in counts (Q1=6, Q3=72). For each top-10 category by total review count, metrics are computed by class: mean stars, mean

review count, and tips-to-reviews ratio.

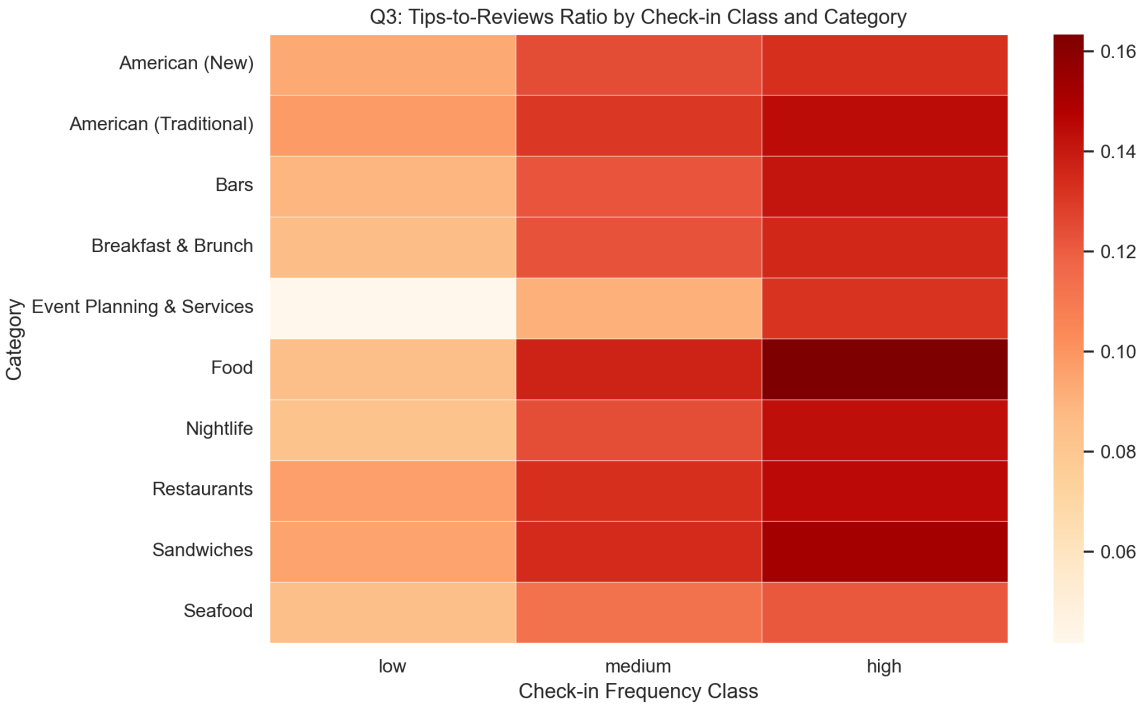

Figure 5: Q3 heatmap: tips-to-reviews ratio by category and check-in class.

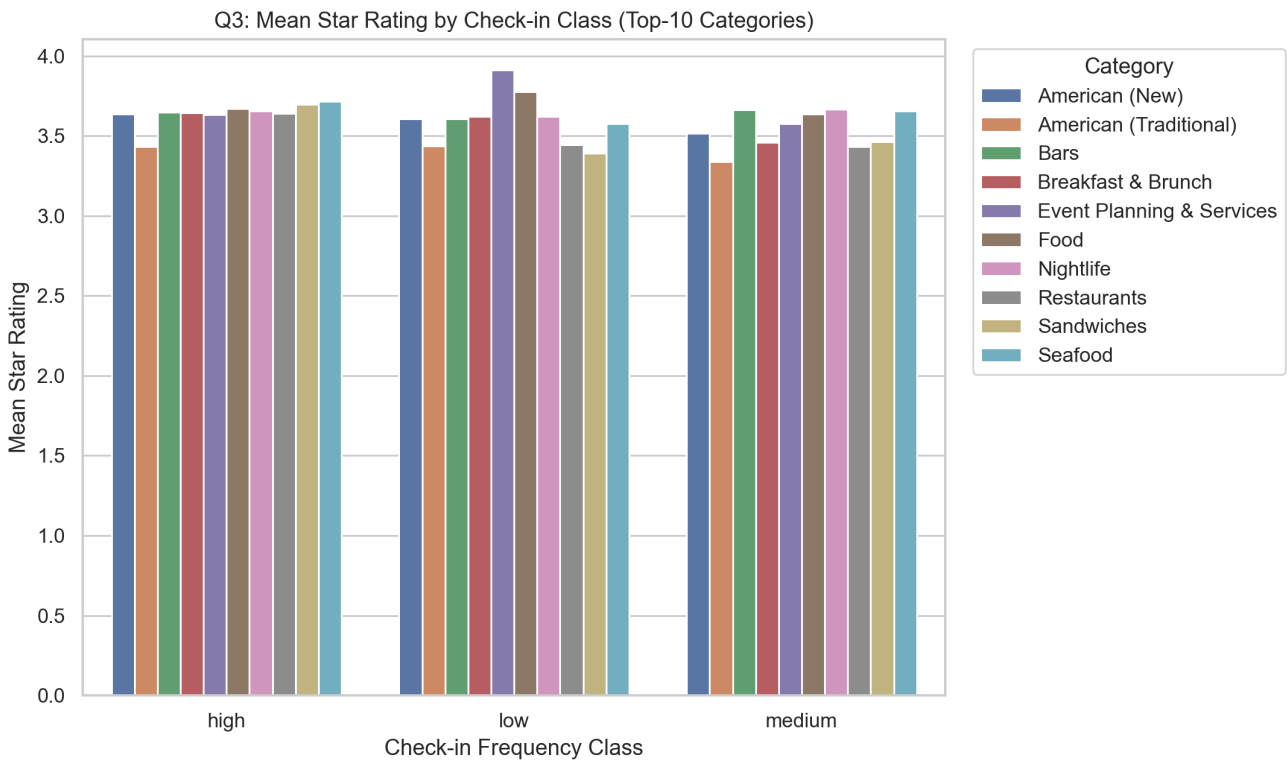

Figure 6: Q3 bar plot: mean stars by check-in class across top categories.

Interpretation:

- High check-in businesses generally have substantially higher mean review volumes.
- Tips-to-reviews intensity is commonly strongest in the high check-in group.

2 Neo4j / Cypher Querying

Workflow and Execution

The full Neo4j Part 2 pipeline is implemented in `queries/p2_neo4j.py`. It writes all Cypher/GDS statements to file, executes each section, and stores outputs/figures in report folders.

Run sequence:

```
source .venv/bin/activate
pip install -r requirements.txt
python queries/p2_neo4j.py
```

Generated artifacts:

- Cypher/GDS statements: `queries/cypher_part2_queries.txt`
- Data outputs: `report/p2/data/n*.csv`, `report/p2/data/n*.json`
- Figures: `report/p2/figures/n*.png`

NQ1: PageRank on User-Business Projection

The query projects a directed weighted graph from users to businesses using review stars as weights and runs GDS PageRank with minimum 20 iterations.

Deliverables produced by script:

- Top-15 businesses by PageRank with review count and average stars
- Spearman correlations: PageRank vs review-count ranking; PageRank vs average-star ranking
- Rank-gap analysis for businesses over/under-ranked by PageRank vs simple metrics

figures/n1_top15_pagerank_bar.png

Figure 7: NQ1: Top 15 businesses by PageRank score.

figures/n1_pagerank_vs_reviewcount_scatter.png

Figure 8: NQ1: PageRank score vs review count.

NQ2: Louvain Community Detection

The script runs Louvain on the `FRIENDS_WITH` social graph, filters communities with at least 25 members, and computes top states, top categories, and geographic concentration index (GCI).

figures/n2_top10_geo_concentration.png

Figure 9: NQ2: Top communities by geographic concentration index.

NQ3: Node Similarity (Jaccard) for Market Saturation

Node Similarity is run on business nodes via shared reviewer neighborhoods. Threshold starts at 5 common reviewers and falls back to 4 or 3 if needed. Results include top-5 most saturated and top-5 most fragmented city-category combinations, excluding combinations with fewer than 5 businesses.

figures/n3_market_saturation_top_bottom.png

Figure 10: NQ3: Most saturated and most fragmented city-category combinations.

The script also computes mean stars, standard deviation of stars, and mean review count for the most and least saturated combinations.

NQ4: Betweenness vs Degree Centrality

Betweenness and degree are computed on the social graph. The top-20 overlap is reported, and high-betweenness-low-degree users are compared against high-degree users on review count, distinct cities reviewed, and distinct categories reviewed.

figures/n4_group_comparison_bar.png

Figure 11: NQ4: Group-level comparison for centrality-defined user sets.

NQ5: Link Prediction Pipeline for Recommendations

The script builds a chronological holdout (each user's latest edge as test), runs a GDS link prediction pipeline, and reports AUC-ROC, Precision@10, top predictive features, and top-3 recommendations for 5 sampled users.

figures/n5_lp_metrics_bar.png

Figure 12: NQ5: Link prediction evaluation metrics.

3 Complete Query Statements

MongoDB Aggregation Pipelines

```
// DSM Assignment 2 Part 2 - MongoDB Querying
// Generated by queries/p2_mongo.py

// ==========================================
// P2-Q1A. Cohort Stats (mean/std stars, length, useful)
// ==========================================

db.reviews.aggregate(
[
    {
        "$lookup": {
            "from": "users",
            "localField": "user_id",
            "foreignField": "_id",
            "as": "u"
        }
    },
    {
        "$unwind": "$u"
    },
    {
        "$match": {
            "u.yelping_since": {
                "$type": "date"
            }
        }
    },
    {
        "$addFields": {
            "cohort_year": {
                "$year": "$u.yelping_since"
```

```
            },
            "review_char_len": {
                "$strLenCP": {
                    "$ifNull": [
                        "$text",
                        ""
                    ]
                }
            }
        }
    },
    {
        "$group": {
            "_id": "$cohort_year",
            "mean_star_rating": {
                "$avg": "$stars"
            },
            "stddev_star_rating": {
                "$stdDevPop": "$stars"
            },
            "mean_review_char_length": {
                "$avg": "$review_char_len"
            },
            "mean_useful_votes_per_review": {
                "$avg": "$useful"
            },
            "review_count": {
                "$sum": 1
            }
        }
    },
    {
        "$sort": {
            "_id": 1
        }
    }
]
);

// ==========================================
// P2-Q1B. Cohort Star Distribution (1-5 proportions)
// ==========================================

db.reviews.aggregate(
[
    {
        "$lookup": {
            "from": "users",
            "localField": "user_id",
            "foreignField": "_id",
            "as": "u"
        }
    },
    {
        "$unwind": "$u"
    },
    {
        "$match": {
            "u.yelping_since": {
                "$type": "date"
            }
        }
    },
    {
        "$addFields": {
            "cohort_year": {
                "$year": "$u.yelping_since"
            },
            "star_int": {
                "$toInt": "$stars"
            }
        }
    },
    {
        "$group": {
            "_id": {
                "cohort_year": "$cohort_year",
```

```
                "star": "$star_int"
            },
            "count": {
                "$sum": 1
            }
        }
    },
    {
        "$group": {
            "_id": "$_id.cohort_year",
            "total_reviews": {
                "$sum": "$count"
            },
            "star_buckets": {
                "$push": {
                    "star": "$_id.star",
                    "count": "$count"
                }
            }
        }
    },
    {
        "$project": {
            "_id": 1,
            "total_reviews": 1,
            "star_buckets": 1
        }
    },
    {
        "$sort": {
            "_id": 1
        }
    }
]
);

// ==========================================
// P2-Q2. Monthly Category Ratings + Trend Consistency Base
// ==========================================

db.reviews.aggregate(
[
    {
        "$lookup": {
            "from": "businesses",
            "localField": "business_id",
            "foreignField": "_id",
            "as": "b"
        }
    },
    {
        "$unwind": "$b"
    },
    {
        "$match": {
            "b.categories": {
                "$type": "array"
            }
        }
    },
    {
        "$unwind": "$b.categories"
    },
    {
        "$group": {
            "_id": {
                "category": "$b.categories",
                "year": {
                    "$year": "$date"
                },
                "month": {
                    "$month": "$date"
                }
            },
            "avg_star": {
                "$avg": "$stars"
            },
```

```
            "review_count": {
                "$sum": 1
            }
        }
    },
    {
        "$sort": {
            "_id.category": 1,
            "_id.year": 1,
            "_id.month": 1
        }
    },
    {
        "$group": {
            "_id": "$_id.category",
            "total_reviews": {
                "$sum": "$review_count"
            },
            "monthly": {
                "$push": {
                    "year": "$_id.year",
                    "month": "$_id.month",
                    "avg_star": "$avg_star",
                    "review_count": "$review_count"
                }
            }
        }
    },
    {
        "$match": {
            "total_reviews": {
                "$gte": 500
            }
        }
    },
    {
        "$sort": {
            "total_reviews": -1
        }
    }
]
);

// ==========================================
// P2-Q3A. Check-in Counts per Business (for quartiles)
// ==========================================

db.checkins.aggregate(
[
    {
        "$project": {
            "business_id": "$_id",
            "checkin_count": {
                "$size": {
                    "$ifNull": [
                        "$dates",
                        []
                    ]
                }
            }
        }
    }
]
);

// ==========================================
// P2-Q3B. Top-10 Categories by Total Review Count
// ==========================================

db.businesses.aggregate(
[
    {
        "$match": {
            "categories": {
                "$type": "array"
            }
        }
```

```
    },
    {
        "$unwind": "$categories"
    },
    {
        "$group": {
            "_id": "$categories",
            "total_review_count": {
                "$sum": "$review_count"
            }
        }
    },
    {
        "$sort": {
            "total_review_count": -1
        }
    },
    {
        "$limit": 10
    }
]
);

// ==========================================
// P2-Q3C. Cross-tab by Check-in Class and Top-10 Categories
// Quartile cutoffs from check-ins: Q1=6.00, Q3=72.00
// ==========================================

db.businesses.aggregate(
[
    {
        "$match": {
            "categories": {
                "$type": "array"
            }
        }
    },
    {
        "$match": {
            "categories": {
                "$in": [
                    "Restaurants",
                    "Food",
                    "Nightlife",
                    "Bars",
                    "American (Traditional)",
                    "American (New)",
                    "Breakfast & Brunch",
                    "Sandwiches",
                    "Seafood",
                    "Event Planning & Services"
                ]
            }
        }
    },
    {
        "$lookup": {
            "from": "checkins",
            "localField": "_id",
            "foreignField": "_id",
            "as": "c"
        }
    },
    {
        "$lookup": {
            "from": "tips",
            "let": {
                "bid": "$_id"
            },
            "pipeline": [
                {
                    "$match": {
                        "$expr": {
                            "$eq": [
                                "$business_id",
                                "$$bid"
                            ]
```

```
                    }
                }
            },
            {
                "$group": {
                    "_id": null,
                    "tip_count": {
                        "$sum": 1
                    }
                }
            }
        ],
        "as": "tip_info"
    }
},
{
    "$addFields": {
        "checkin_count": {
            "$ifNull": [
                {
                    "$let": {
                        "vars": {
                            "c1": {
                                "$arrayElemAt": [
                                    "$c",
                                    0
                                ]
                            }
                        },
                        "in": {
                            "$size": {
                                "$ifNull": [
                                    "$$c1.dates",
                                    []
                                ]
                            }
                        }
                    }
                },
                0
            ]
        },
        "tip_count": {
            "$ifNull": [
                {
                    "$arrayElemAt": [
                        "$tip_info.tip_count",
                        0
                    ]
                },
                0
            ]
        }
    }
},
{
    "$addFields": {
        "checkin_class": {
            "$switch": {
                "branches": [
                    {
                        "case": {
                            "$lte": [
                                "$checkin_count",
                                6.0
                            ]
                        },
                        "then": "low"
                    },
                    {
                        "case": {
                            "$gt": [
                                "$checkin_count",
                                72.0
                            ]
                        },
                        "then": "high"
```

```
                    }
                ],
                "default": "medium"
            }
        }
    }
},
{
    "$unwind": "$categories"
},
{
    "$match": {
        "categories": {
            "$in": [
                "Restaurants",
                "Food",
                "Nightlife",
                "Bars",
                "American (Traditional)",
                "American (New)",
                "Breakfast & Brunch",
                "Sandwiches",
                "Seafood",
                "Event Planning & Services"
            ]
        }
    }
},
{
    "$group": {
        "_id": {
            "category": "$categories",
            "checkin_class": "$checkin_class"
        },
        "business_count": {
            "$sum": 1
        },
        "mean_star_rating": {
            "$avg": "$stars"
        },
        "mean_review_count": {
            "$avg": "$review_count"
        },
        "total_tips": {
            "$sum": "$tip_count"
        },
        "total_reviews": {
            "$sum": "$review_count"
        }
    }
},
{
    "$project": {
        "_id": 0,
        "category": "$_id.category",
        "checkin_class": "$_id.checkin_class",
        "business_count": 1,
        "mean_star_rating": 1,
        "mean_review_count": 1,
        "tips_to_reviews_ratio": {
            "$cond": [
                {
                    "$gt": [
                        "$total_reviews",
                        0
                    ]
                },
                {
                    "$divide": [
                        "$total_tips",
                        "$total_reviews"
                    ]
                },
                0
            ]
        }
    }
```

```
    },
    {
        "$sort": {
            "category": 1,
            "checkin_class": 1
        }
    }
]
);
```

Neo4j Cypher and GDS Statements

4 Artifact Index

- Mongo script: `queries/p2_mongo.py`
- Neo4j script: `queries/p2_neo4j.py`
- Mongo queries: `queries/mongo_part2_queries.txt`
- Neo4j queries: `queries/cypher_part2_queries.txt`
- Data outputs: `report/p2/data/`
- Figures: `report/p2/figures/`